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Ocean tools-

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Project Abyssal Trust

Cisco Consulting Proposal for The Metals Company (TMC)

Bat Bridge Partners — Cisco HS Capstone 2026

PART 1: THE PROBLEMS

Before any solution, here is an honest inventory of what is actually wrong at TMC. This ordering matters: judges reward teams that diagnose before they prescribe.

Problem 1 — The Trust Deficit (the root problem)

TMC's single greatest business risk is not technical. It is that **nobody believes their environmental claims**. Their pre-feasibility study was written largely by their own employees rather than independent experts. Their "zero waste" claim excludes the sediment-laden water they discharge back into the ocean. Greenpeace, the Environmental Justice Foundation, MiningWatch, and the International Seabed Authority itself have all publicly attacked their credibility. When your regulator, your critics, and independent analysts all distrust your data, every permit, every investor dollar, and every insurance policy gets harder and more expensive.

Key insight our solution is built on: TMC does not have a mining problem. It has a *proof* problem.

Problem 2 — 1960s Collection Technology

TMC's collector vehicles are conventional: tracked machines and suction systems whose basic architecture has barely changed in sixty years. They cannot distinguish a bare nodule from a nodule hosting living organisms, and they generate large sediment plumes — the two specific harms critics cite most. This is simultaneously their biggest ethical liability and our biggest opportunity, because it means the AI maturity of their core operation is near zero. Everything we propose is additive. Nothing is ripped out.

Problem 3 — An Unsecured, Geopolitically Exposed Operation

TMC operates in the Clarion-Clipperton Zone under a U.S. permit that the ISA calls a violation of international law — meaning multiple nation-states and activist groups have motive to disrupt, discredit, or surveil the operation. Their attack surface is enormous and almost entirely

operational technology (OT): a production vessel functioning as a floating industrial plant, remotely operated and autonomous underwater vehicles, a satellite link to shore, and an environmental sensor network whose data feeds directly into regulatory decisions.

The nightmare scenarios are specific: a hostile actor spoofs or corrupts environmental monitoring data (either to trigger a false violation that kills the license, or to hide a real one and destroy TMC's credibility when discovered); ransomware locks the vessel's industrial controls mid-operation, hundreds of miles from help; or vessel-tracking (AIS) manipulation near the operating area creates collision risk or false narratives. The Baltic Sea taught the world in 2024–25 that maritime infrastructure incidents are cheap to cause and nearly impossible to attribute. TMC is a higher-profile, more hated target than a telecom cable.

Problem 4 — Financial Fragility Constrains Everything

TMC posted a net loss of roughly \$320M in 2025 and independent analysts question whether the project can ever be profitable. Any solution costing "whatever it takes" is dead on arrival. The proposal must be phased, must lead with the components that unlock revenue and licensing (not vanity tech), and must show a defensible return story.

Problem 5 — Regulatory Dependency on Data Integrity

NOAA's license comes with environmental compliance conditions. In this industry, the environmental data pipeline IS the license. If that pipeline is not provably trustworthy — tamper-evident, auditable, chain-of-custody intact — then a single disputed dataset can suspend operations. Today, no deep-sea miner treats environmental telemetry as security-critical infrastructure. That is the gap.

PART 2: THE SOLUTION — "THE GLASS OCEAN"

The Big Idea (our differentiator)

Every other team will propose AI + firewalls to make their company more efficient. We propose something different: **turn TMC's environmental monitoring into a radically transparent, cryptographically verifiable public utility — the Glass Ocean — secured end-to-end by Cisco infrastructure.**

Instead of TMC saying "trust us," the seafloor broadcasts its own evidence. Environmental sensor data is signed at the source, transported over a secured Cisco OT network, logged immutably, and exposed on a public dashboard that regulators, scientists, and yes, even

Greenpeace can watch in near-real-time. TMC's biggest liability (distrust) becomes its moat: the first extraction company whose environmental claims are independently verifiable by design.

This reframes the entire project: security isn't a cost center protecting mining. **Security is the product that makes the mining licensable.**

AI Use Case 1 — Selective Harvesting Retrofit ("Eyes on the Collector")

Computer-vision perception retrofitted onto TMC's existing collector vehicles. Cameras and lights feed an onboard AI model that classifies nodules in the vehicle's path in real time: bare nodule (harvest) versus nodule with visible epifauna — sponges, corals, xenophyophores (skip). The collector head adjusts pickup on the fly. A second model monitors the sediment plume live via turbidity sensors and optical backscatter, automatically throttling vehicle speed when plume density exceeds a threshold.

Why this is additive, not replacement: the vehicles, the ship, the riser system all stay. We are adding a perception layer to machinery that currently operates blind — the industrial equivalent of adding driver-assist to an existing truck fleet. (We deliberately cite Impossible Metals' selective-harvesting robots as proof the concept works, then differentiate: they built a new robot; we retrofit intelligence onto proven heavy equipment a real company already owns.)

Where the AI runs: on the vessel, at the edge. A shipboard Cisco UCS / AI POD-class compute node does inference locally because (a) there is nowhere near enough satellite bandwidth to stream 4K seafloor video to a cloud, and (b) a 700ms+ satellite round-trip is useless for a vehicle decision that must happen in tens of milliseconds. Sustainability pillar bonus: edge inference slashes satellite bandwidth (energy + cost) and lets us use pre-trained vision backbones fine-tuned on nodule imagery rather than training from scratch.

AI Use Case 2 — The Environmental Sentinel Network ("The Ocean's Black Box")

A distributed AI monitoring system that is deliberately independent from the mining control system: moored sensor strings and monitoring AUVs measuring turbidity, sediment plume drift, bioacoustics (marine mammal calls, ambient soundscape), water chemistry, and periodic photographic biodiversity transects. Onboard AI does three jobs: plume-drift prediction (a physics-informed model forecasting where sediment travels over 6–48 hours so operations can be adjusted *before* a violation, not after), bioacoustic event detection (whale presence auto-triggers operational quieting protocols), and anomaly detection on the sensors themselves (an AI watching for sensor drift, gaps, or statistically implausible readings — i.e., tampering or failure).

This second use case is what feeds the Glass Ocean. And critically for the ethics rubric: the Sentinel Network's data is treated as *evidence*, engineered to a chain-of-custody standard, not as internal telemetry.

The Cisco Stack (front to back)

On the seafloor and in the water column: sensor pods and AUVs are, in network terms, untrusted industrial endpoints. Their data lands at the vessel.

Honest engineering note we volunteer before judges catch it: radio waves do not propagate underwater. Subsea links use acoustic modems, tethers, or optical links — none of which are Cisco products. Cisco's role begins the moment data reaches a surface asset. Claiming otherwise would be the "space-hardened router" mistake, and we refuse to make it.

On the surface: Cisco Ultra-Reliable Wireless Backhaul (CURWB) connects the production vessel to its support craft, surface buoys (the tops of moored sensor strings), and USVs — this is CURWB's actual designed use case (ports, mining, moving industrial assets). Onboard the vessel, Cisco Catalyst Industrial Ethernet switches form the ship's OT backbone.

Segmentation — the architectural heart: Cisco Identity Services Engine (ISE) plus industrial firewalling divides the vessel into strict zones: (1) vessel IT / crew network, (2) mining OT (collector controls, riser, dynamic positioning), (3) the Environmental Sentinel network, and (4) the satellite uplink DMZ. The Sentinel zone is one-way-out by design (data diode pattern): environmental data can leave, but nothing on the ship — including TMC's own operators — can write into it. That single design choice is what makes "the miner can't fake its own environmental data" a *provable architectural claim* instead of a policy promise. Cisco Cyber Vision passively inventories and monitors every OT device in zones 2 and 3, flagging anomalous industrial commands (the Ascension lesson: 7 compromised servers took down a 25,000-server network because nothing was segmented; here, a breached crew laptop can never reach a collector PLC).

Identity and access: Cisco Duo enforces phishing-resistant MFA for every remote operations session and every shore-side engineer — directly closing the MGM lesson (one social-engineered helpdesk call) and the Colonial lesson (one password, no MFA). Shore-based remote access flows through Cisco Secure Access (ZTNA), not a flat VPN.

Ship-to-shore and visibility: Cisco ThousandEyes monitors the satellite path and shore connectivity end-to-end — when the link degrades mid-Pacific, operators know in seconds whether the fault is shipboard, satellite, or shore. Cisco Umbrella provides DNS-layer protection for all shipboard internet egress (the Ascension entry vector: one malicious download). Cisco XDR correlates detections across endpoint, network, and identity; **Splunk (a Cisco company — a point most teams won't know) serves as the immutable, timestamped audit log** for both security events and the environmental data ledger.

The Glass Ocean layer: every Sentinel reading is hash-chained and signed as it enters Splunk, producing a tamper-evident ledger. A public dashboard publishes near-real-time plume, acoustic, and biodiversity indicators, with a cryptographic proof page that lets any third party verify the data hasn't been altered since capture. A standing Webex-based "regulator observation room" gives NOAA and invited independent scientists live windows into operations — turning oversight from an adversarial audit into a continuous feed. (Webex is a deliberately unglamorous inclusion that scores "use of Cisco technologies" breadth honestly.)

Value & Success Metrics (the pillar TMC's finances make hardest — so we quantify around trust, not tonnage)

The honest framing: our solution does not make a marginal mine profitable. It makes an unlicensable operation licensable and an uninsurable operation insurable. Concrete metrics: percentage of fauna-hosting nodules avoided (target from pilot data, e.g., >90% detection recall); plume-threshold exceedance events per 1,000 operating hours (target: trending to zero with predictive throttling); mean time to detect an OT anomaly (from "never/unknown" to minutes via Cyber Vision + XDR); environmental data disputes resolvable by ledger proof (target: 100%); avoided downtime cost (using CommonSpirit's ~\$160M and Ascension's ~\$1.1B as sector analogies for what one extended OT outage costs a mid-cap operator — an offshore vessel idled by ransomware burns charter, crew, and schedule costs daily); and unlocked capital — ESG-screened investors and insurers currently priced out by unverifiable claims. One honest caveat we state on the slide: several of these are risk-avoidance and optionality values, not guaranteed revenue, and we defend that openly rather than inventing fake ROI percentages.

Sustainability & Deployment (rubric pillar 4)

Edge-first inference on shipboard UCS (bandwidth and energy efficiency; the satellite link is the scarcest resource in the architecture). Pre-trained vision models fine-tuned on nodule imagery, not trained from scratch. Hybrid hosting: heavy model retraining happens onshore between voyages; only lightweight model updates ship over the link. Hardware is containerized and modular so the same stack redeploys across future vessels — and, in a licensing twist that addresses TMC's cash problem, the Glass Ocean architecture itself is packageable as a compliance product TMC could license to other seabed operators, converting a cost center into a potential revenue line.

Implementation Roadmap (phased for a cash-poor client)

Phase 0 (months 0–3): security fundamentals only — Duo, Umbrella, segmentation design, Cyber Vision passive discovery. Lowest cost, closes the MGM/Colonial/Ascension class of failure immediately. Phase 1 (months 3–9): vessel OT segmentation implemented; Sentinel network pilot with hash-chained logging; ThousandEyes on the uplink. Phase 2 (months 9–18):

selective-harvesting vision retrofit on one collector vehicle; plume-prediction model in advisory (human-in-the-loop) mode. Phase 3 (months 18–30): Glass Ocean public dashboard and regulator observation room; vision system moves from advisory to automatic; fleet scale-out. Each phase has an exit ramp — deliberately, because the client's financing is uncertain, and a roadmap that pretends otherwise would be fantasy.

PART 3: FRONT-TO-BACK AUDIT (attacking our own solution before the judges do)

"Why would a company losing \$320M/year buy any of this?" Strongest objection, so it's first. Answer: Phase 0 costs a rounding error against one incident, and Phases 2–3 are gated on financing milestones. Deeper answer: TMC's constraint on raising money IS the trust deficit; the Glass Ocean is pitched as the thing that unlocks capital, not a luxury purchased after capital arrives. If judges still push: we note the architecture is client-portable to any seabed operator (AOMC/Odyssey, national programs), so the design's value doesn't die with TMC's balance sheet.

"Isn't consulting for the ocean's most-protested company unethical?" We answer head-on: the capstone brief says the client wants to modernize "safely & ethically" — our proposal is the mechanism that forces that, since a one-way data architecture removes TMC's ability to lie. If TMC operates under Glass Ocean, the world gets the first verifiable evidence base about deep-sea mining impacts; if the data shows unacceptable harm, that evidence empowers regulators to stop it. We are not defending seabed mining; we are making it impossible to do in the dark. (We should also acknowledge the counterargument — transparency can legitimize an activity that shouldn't happen at all — and say the team debated it. Judges reward visible ethical reasoning over tidy answers.)

"Underwater comms aren't Cisco — doesn't that hole the story?" No, because we scoped it honestly: Cisco owns surface-and-up plus everything shipboard and shoreside, which is where every one of the named failure modes (ransomware, spoofed data, stolen credentials, flat networks) actually lives. Volunteering the limitation is itself a Technical Feasibility point.

"Can computer vision really run on a collector at 4,000m depth?" Cameras, LED lighting, and edge inference on subsea vehicles are established (Impossible Metals does classification-grade vision on nodules today; offshore inspection ROVs run CV routinely). Our retrofit claim is deliberately conservative: advisory mode first, human-in-the-loop, automatic control only in Phase 3. We do not promise a percentage improvement we can't source.

"The Glass Ocean could backfire — real-time data might hand critics ammunition."

Correct, and that's the point; we say so. A company only adopts this if it intends to operate within limits. The predictive-throttling design exists precisely so TMC sees a violation coming before the public feed shows one. Risk remains that leadership rejects transparency — which is a client-adoption risk, not a design flaw, and we flag it as the project's #1 non-technical risk.

"AIS spoofing and hostile vessels — you mentioned the threat, where's the counter?"

Scoped honestly: shipboard radar/AIS cross-checking with anomaly detection is feasible and included in the Sentinel anomaly layer; actual interdiction of hostile vessels is a state function, not a Cisco function. We recommend TMC formalize information-sharing with NOAA and naval authorities — the Baltic Sentry lesson.

Remaining weaknesses we accept: no historical breach at TMC means our impact numbers borrow from healthcare/energy analogies (we label them as analogies); public sentiment risk is real and some judges may simply dislike the client; and the Glass Ocean's originality cuts both ways — it invites more scrutiny than a generic SOC-and-firewalls proposal would. We take that trade. The rubric pays 30 points for originality and this is the most original defensible position on the board.

One-slide summary (for the team)

Client: The Metals Company. **Diagnosis:** a proof problem wearing a mining costume.

Prescription: retrofit intelligence onto blind machines (AI #1), build an incorruptible environmental witness (AI #2), and secure it all with a segmented, zero-trust Cisco architecture whose signature feature — the one-way Glass Ocean — turns the company's worst liability into its license to operate.

This is a genuinely good instinct — you're describing a **platform play**: the Glass Ocean stops being TMC's compliance tool and becomes industry infrastructure. That's exactly what "scalability" means to a Cisco judge. But I need to push back hard on one sentence before it infects your deck, then I'll show you how to structure both ideas so they strengthen rather than dilute.

The pushback first: kill the phrase "take the data and do almost anything with it." Never say anything like that in the room. Your entire core solution is built on the claim that this data is a *trust instrument* — tamper-proof, governed, evidence-grade. The moment you say "we can use the data any way that benefits us," you've contradicted your own thesis in front of judges, and the ethics pillar collapses. Worse, "monetize environmental data however we want" is precisely the behavior (Kaiser's tracking-pixel breach) that we cited as a cautionary tale earlier. The correct framing — and it's *more* impressive, not less — is **governed data products**: specific, named uses under a published data-governance policy. Same ambition, defensible packaging.

Now the two satellite solutions, structured properly:

Ring 1 — "SafeDepth": Deep-Sea Health & Safety Platform. Extend the same shipboard architecture to human safety: connected crew wearables and vessel sensors (man-overboard detection, gas/atmosphere monitoring in enclosed spaces, fatigue-shift analytics for 24/7 remote operations, emergency comms redundancy over the CURWB mesh), all riding the network zones you already built — no new infrastructure, just a new tenant on it. The scalability sentence writes itself: *every* offshore operator (mining, wind, oil & gas, cable repair, research vessels) has identical crew-safety obligations, so SafeDepth is sellable to the whole maritime industry regardless of whether deep-sea mining itself survives. That's a subtle but powerful hedge: your solution outlives your controversial client.

Ring 2 — "Abyssal Weather": Ocean Data as Public Infrastructure. Your Sentinel Network is already measuring currents, temperature, turbidity, and acoustics at 4,000m in one of the least-instrumented regions on Earth. The deep Pacific is a data desert — oceanographers and climate modelers would genuinely value continuous abyssal time-series. So: feed governed, anonymized-of-operations oceanographic data to NOAA, universities, and climate models; use current-prediction internally to optimize vessel positioning, plume forecasting (this directly *boosts* your core AI Use Case 2 — better current data = better plume prediction = fewer violations), and weather-window planning that cuts fuel burn (sustainability pillar). Commercially, ocean-current and weather data has real buyers — shipping route optimization, offshore energy — as a data-as-a-service line under the governance policy. Frame it as "the first commercial abyssal observation network — mining pays for the sensors, science gets the data."

How to present this without diluting the core: use a concentric-rings slide. Center: Glass Ocean + two AI use cases (TMC-specific). Ring 1: SafeDepth (same architecture, new safety tenant, industry-wide market). Ring 2: Abyssal Weather (same sensors, new data products, societal benefit + revenue hedge). One slide, 60 seconds of talk time, and it hits Scalability,

Sustainability, *and* Long-term Benefits simultaneously. Do **not** give the rings more than that — your 100 points live in the core solution, and teams that sprawl across three half-built ideas score worse than teams with one deep idea and two crisp extensions. The rings exist to answer "how does this scale?" before anyone asks.

One more judge-proofing note: Ring 2 finally gives you a clean answer to the "broke client" objection — the sensor network becomes a revenue-generating asset rather than pure compliance cost, and it's the part of the business that's valuable *even if TMC never mines a single nodule commercially*.

Notes

ADVAITH - ARUSHI

Problem: Outdated Mining Technology & Environmental Compliance

TMC's current mining vehicles cannot distinguish marine life from mineral nodules, causing unnecessary disturbance to deep-sea ecosystems. At the same time, regulators require accurate environmental data to prove mining operations remain within environmental limits. If this data is inaccurate, manipulated, or incomplete, TMC could lose regulatory approval.

Solution: AI-Powered Selective Harvesting + Environmental Sentinel Network

Retrofit TMC's existing mining vehicles with **AI-powered computer vision** that identifies marine organisms and selectively collects only safe nodules, reducing habitat disturbance and sediment plumes. Surrounding the mining operation, an **Environmental Sentinel Network** of underwater sensors continuously monitors water quality, sediment movement, ocean currents, biodiversity, and marine mammals.

AI analyzes both the mining vehicles and environmental sensors in real time to:

- Identify marine organisms and avoid sensitive habitats.
- Predict sediment plume movement before environmental thresholds are exceeded.
- Detects marine mammals and recommends slowing or pausing operations.
- Identify faulty or tampered sensors.
- Automatically alert operators when environmental conditions require action.

Verified environmental data is then published to **Glass Ocean**, creating a transparent record for regulators, scientists, and the public.

Cisco Technologies

- **Cisco UCS / AI Infrastructure** runs AI models onboard the vessel for real-time decision-making.
- **Cisco Industrial Ethernet** securely connects cameras, sensors, and mining equipment.
- **Cisco Cyber Vision** continuously monitors operational technology devices and detects abnormal behavior or compromised sensors.
- **Cisco Splunk** creates a tamper-evident audit trail, preserving the chain of custody for all environmental data.
- **Cisco XDR** detects cyber threats or attempts to manipulate environmental data.
- **Cisco ThousandEyes** monitors communication between the vessel and shore, ensuring environmental data reaches regulators reliably.

TMC Capstone — Simplified Overview

Bat Bridge Partners — Cisco HS Capstone 2026

1. Trust Deficit

Snikitha and Harini

Problem: People don't trust TMC's environmental claims, making permits and investment difficult. Their studies were written by their own employees, critics dispute their "zero waste" claim, and regulators, analysts, and environmental groups all doubt their data. Distrust makes capital expensive — it's the root of their money problems too.

Solution: Glass Ocean — a public, tamper-proof environmental monitoring platform. AI collects data from underwater sensors, and the data is digitally verified so regulators and scientists can confirm it hasn't been changed.

Instead of saying "trust us," TMC lets the ocean prove it:

- Sensors on the seafloor collect environmental data.
- That data is protected with Cisco cybersecurity, checked by AI, stored so nobody can secretly change it (not even TMC), and shared on a public dashboard for regulators and scientists.
- Anyone can verify the data is authentic.

This turns trust into evidence.

Ideas to expand the Glass Ocean website demo: a live-style public dashboard (plume levels, whale detections, "harvest decisions today"), a "verify this data" button that shows the tamper-proof check happening, a regulator view vs. public view toggle, and a data-governance page showing exactly who can access what.

Cisco Technologies Used:

- **Splunk (owned by Cisco):** the tamper-proof record book — every sensor reading is time-stamped and hash-chained the moment it arrives, so any change would be instantly detectable. This is the engine behind the "verify this data" feature.
- **Cisco Catalyst Industrial Ethernet switches:** the rugged shipboard network that carries sensor data from the moment it reaches the vessel — built for harsh industrial environments, not office buildings.

- **Cisco ThousandEyes:** watches the satellite connection between ship and shore end-to-end, so if the public dashboard ever goes dark, TMC can prove in seconds whether it was a technical fault or something suspicious.
- **Webex:** powers the "regulator observation room" — a standing live window where NOAA and independent scientists can watch operations in real time instead of auditing paperwork months later.

Main Category: Value + Ethics **Demo:** Website

2. Outdated Mining Technology

ARUSHI

Problem: Current collector vehicles are "blind" and disturb marine life and sediment. The machines can't tell a bare rock from one hosting living creatures, and they can't prove they operated cleanly even when they did — no trustworthy record exists at the exact moment regulators demand one.

Solution: AI Selective Harvesting — add cameras and computer vision to existing mining vehicles so they identify marine life, skip sensitive areas, and slow down when sediment plumes become too large. Every decision the vehicle makes (creature avoided, speed reduced) is automatically recorded into the same tamper-proof system as the Glass Ocean — so the machines get **eyes** (they can avoid harm) and a **voice** (they can prove it).

Nothing gets replaced — it's a retrofit onto equipment TMC already owns, like adding driver-assist to an existing truck.

Cisco Technologies Used:

- **Cisco UCS servers (AI POD-class):** the shipboard "brain" — the AI vision models run on this computer aboard the vessel itself, because there isn't enough satellite bandwidth to send seafloor video to the cloud, and decisions must happen in milliseconds.
- **Cisco Ultra-Reliable Wireless Backhaul (CURWB):** the wireless link connecting the main vessel to support boats, surface buoys, and drones — designed exactly for moving industrial assets like ports and mines.
- **Cisco Catalyst Industrial Ethernet:** carries the collector's camera feeds and decision logs across the ship's network reliably in an industrial environment.

(Honest note we volunteer in the presentation: the underwater leg itself — vehicle to surface — uses acoustic/tethered links that aren't Cisco products. Cisco takes over the moment data reaches a surface asset. Saying this before judges ask earns feasibility credibility.)

Main Category: Technical + Sustainability **Demo:** Model

Website tie-in idea: a first-person "robot cam" view on the site — see through the collector's eyes as it classifies rocks in real time (green box = safe to collect, red box = life detected, skip). That single visual explains the entire AI use case in five seconds.

3. Cybersecurity & Operational Risks

Problem: Ships, robots, sensors, and satellite links could be hacked or manipulated, disrupting operations or altering environmental data. TMC is a high-profile target — nation-states and activist groups have motives, and a single attack could fake an environmental violation (killing the license) or lock up the ship's controls mid-ocean.

Solution: Zero Trust Security Architecture — secure identities, segment operational networks, monitor industrial devices, and protect environmental data from tampering using Cisco security technologies:

- **Duo:** every person proves who they are (stops the phone-scam attacks that took down MGM).
- **Network segmentation (ISE):** the ship is divided into isolated zones — a hacked crew laptop can never reach the mining controls (the lesson from the Ascension hospital attack).
- **Cyber Vision:** watches every industrial device for strange behavior.
- **One-way data design:** environmental data can flow out to the public, but nobody — not even TMC — can reach in and edit it.

Security isn't just protection here — it's what makes the whole Glass Ocean trustworthy in the first place.

Cisco Technologies Used:

- **Cisco Duo:** phishing-resistant multi-factor authentication for every person and every remote session — the direct fix for the "one phone call took down MGM" and "one password shut down Colonial Pipeline" class of attack.
- **Cisco ISE (Identity Services Engine):** enforces the ship's network zones — crew network, mining controls, environmental sensors, and satellite link are walled off from each other, so one infected laptop can never touch a mining machine.
- **Cisco Cyber Vision:** passively watches every industrial device (robots, sensors, controllers) and flags commands that don't look normal — like a security camera for machine behavior.

- **Cisco Umbrella:** blocks dangerous websites and downloads at the DNS level for everyone aboard — the exact entry point (one malicious download) that started the Ascension hospital attack.
- **Cisco Secure Access (ZTNA):** shore-based engineers connect through zero-trust access that verifies every session, replacing the old-style flat VPN.
- **Cisco XDR:** the correlation brain — connects signals from endpoints, network, and identity so one suspicious event anywhere becomes a visible alert everywhere.

Main Category: Security

One-line summary

Give the machines eyes (AI harvesting), give the ocean a voice (Glass Ocean), and lock both behind doors nobody can pick (Zero Trust) — turning TMC's biggest liability into its license to operate.

Potential Questions

Category 1: The "Enabler" Trap

Question 1: *"Even if the public dashboard shows TMC is destroying a reef in real-time, the data doesn't stop the drill. How does simply watching the destruction happen make your solution ethical?"*

- **The Counter-Argument:** The network does more than just watch. It enforces smart contracts and automatic operational boundaries.
- **The Script:**
"Our solution does not just watch; it acts as a digital kill-switch. The Cisco network architecture is integrated directly into the ship's Operational Technology (OT). This means environmental thresholds are hardcoded into the system. If the Sentinel Network detects an unauthorized ecological breach, it doesn't just send an alert to a dashboard—it automatically triggers a network-level override that throttles or halts the collector vehicle's propulsion. We aren't just broadcasting the destruction; we are automating the brakes."

Question 2: *"Cisco prides itself on CSR. By making an environmentally devastating operation 'licensable,' isn't Cisco profiting from the destruction of the deep sea?"*

- **The Counter-Argument:** Cisco's mission is to secure critical infrastructure. If this industry is going to happen, leaving it unsecured is an environmental crime in itself.
- **The Script:**
"Cisco's corporate responsibility is built on the belief that secure, transparent infrastructure prevents global catastrophes. The reality is that multiple nation-states are going to mine the seabed, with or without Cisco. If Cisco steps away, this industry will be built on opaque, unsecured, proprietary networks where ecocide can be easily hidden. By securing this infrastructure, Cisco isn't endorsing mining; Cisco is ensuring that the global commons cannot be exploited in total darkness. The most unethical thing Cisco could do is let a high-risk industry operate without the world's best security."

Question 3: *"What if the regulator (NOAA or ISA) chooses to look the other way because of political pressure? What value does your data have then?"*

- **The Counter-Argument:** You are bypassing the regulators and giving the data directly to the court of public opinion.

- **The Script:**

"That is exactly why the Glass Ocean is designed to bypass any single point of political failure. By publishing cryptographically signed, un-falsifiable data to an open-source, public utility dashboard, we take the exclusive power away from backroom political regulators. If NOAA chooses to look the other way, they have to do it while Greenpeace, international courts, and the global media are looking at the exact same live pollution data. The data becomes a tool for public and legal litigation that no regulator can ignore."

Category 2: The AI & "Greenwashing" Trap

Question 4: *"Deep-sea microbes are microscopic. Aren't you just teaching your AI to hide the visible destruction while obliterating foundational microscopic life?"*

- **The Counter-Argument:** Microscopic life is tied to macroscopic habitats. Protecting the visible structures protects the invisible ecosystem.
- **The Script:**
"We recognize that the microscopic ecosystem is the foundation of the deep sea. However, deep-sea microbes and megafauna do not exist in isolation—they are symbiotic. Microscopic life relies on the exact same hard nodule substrates and biological structures as the visible sponges and corals. By training our AI to detect and skip nodules hosting visible life, we are explicitly preserving the complex biological hotspots where microbial diversity is highest. We are using visible indicators as a shield for the invisible ecosystem."

Question 5: *"If your AI automatically slows down the vehicle to stay just under the legal pollution limit, isn't that automated greenwashing?"*

- **The Counter-Argument:** Keeping operations strictly within legal limits is how environmental regulations are successfully enforced in every land-based industry.
- **The Script:**
"Enforcing compliance is not greenwashing; it is the literal definition of environmental protection. On land, factory smokestacks use scrubbers to keep emissions under legal limits. Our AI plume throttle does the exact same thing underwater. If a legal limit is set by scientists to protect the water column, forcing the machinery to dynamically respect that limit prevents ecological collapse. It

shifts the industry from an uncontrolled, blind extraction model to a strictly bounded, highly regulated operation."

Flaw

Flaws of glass ocean solution

Flaw	Cisco Solution
Someone tampers with a sensor.	Cyber Vision detects unusual sensor behavior; XDR correlates it with other events; Splunk preserves the audit trail.
Someone steals employee credentials.	Duo enforces phishing-resistant multi-factor authentication before anyone can access Glass Ocean.
Hackers move laterally through the ship's network.	ISE and Secure Firewall segment the network so mining controls, environmental sensors, and crew systems remain isolated.
Satellite connection fails.	ThousandEyes identifies whether the issue is on the vessel, satellite link, or shore. AI continues processing locally until connectivity returns.
Public doesn't trust company-owned sensors.	Publish signed audit logs from Splunk , allow regulator access via Webex , and support independent sensor deployments.
Too much environmental data overwhelms users.	AI summarizes trends into simple compliance reports and dashboards while keeping detailed data available for experts.

Flaw	Why it's a Problem	Possible Solution
1. "How do we know TMC isn't manipulating the sensors before the data is uploaded?"	Even if the data can't be changed after it's stored, someone could tamper with the sensor itself.	Use multiple independent sensors measuring the same thing. AI cross-checks readings and flags disagreements. Add periodic inspections by independent scientists or AUVs.
2. Public dashboard exposes sensitive business information.	Competitors or activists could see mining locations and operational details.	Publish environmental data only , while keeping operational details private. Show summarized locations instead of exact coordinates.
3. AI can make mistakes identifying marine life.	False negatives could still harm ecosystems.	Keep humans in the loop initially. AI recommends actions, but operators approve them until accuracy is proven. Regularly retrain models with new data.

4. The system is expensive.	TMC already has financial challenges.	Deploy in phases. Start with one mining vehicle and one monitoring zone before expanding.
5. Regulators still may not trust company-owned sensors.	They may argue that TMC controls the hardware.	Invite universities or regulators to deploy some of the sensors themselves. Create a shared governance model where not all equipment is company-owned.
6. Satellite communication is unreliable.	Ships operate far offshore and can lose connectivity.	Process AI locally (edge computing) and sync data once the connection returns. This also reduces bandwidth costs.
7. Cyberattacks could disable monitoring.	If sensors go offline, transparency disappears.	Add redundant sensors, backup communication paths, and continuous monitoring. AI alerts operators when sensors fail or behave abnormally.
8. Transparency could hurt TMC.	If the dashboard shows environmental problems, critics will use that information.	That's actually part of the value. AI provides early warnings , allowing operators to slow or stop mining before thresholds are exceeded. Transparency becomes evidence of responsible operation rather than a weakness.
9. Environmental impacts are still uncertain.	Even perfect monitoring doesn't prove deep-sea mining is safe.	Position Glass Ocean as a decision-support system. If the data shows unacceptable impacts, operations should be adjusted or stopped. You're improving evidence, not guaranteeing mining is harmless.

glass ocean

TEAM BRIEF 1 of 3 — GLASS OCEAN (Trust Deficit)

For the two presenters owning Problem/Solution 1

Project Glass Ocean — Bat Bridge Partners — Cisco HS Capstone 2026

The whole project in 90 seconds (same in all three briefs)

Our client is **The Metals Company (TMC)** — a real, publicly traded company trying to become the first commercial deep-sea miner. They harvest potato-sized rocks (polymetallic nodules) from the Pacific seafloor at 4,000m depth; the rocks are rich in nickel, cobalt, copper, and manganese — the metals batteries and data centers need. TMC is betting its future on AI-driven modernization to make this viable, which is exactly the client the capstone prompt describes: excited about AI, terrified of the security and ethics risks that come with it.

Our proposal has three linked parts: (1) **Glass Ocean** — a tamper-proof, publicly verifiable environmental monitoring platform (your part), (2) **AI Selective Harvesting** — computer vision retrofitted onto their mining vehicles so they avoid sea life, and (3) **Zero Trust Security** — the Cisco architecture protecting it all. The one-line thesis that ties them: *give the machines eyes, give the ocean a voice, and lock both behind doors nobody can pick.*

Scalability extensions (mention only if asked): SafeDepth (crew health/safety platform reusing our network) and Abyssal Weather (selling/sharing ocean-current data with science and shipping).

Your problem: the Trust Deficit

TMC's core business risk isn't technical — it's that **nobody believes them**. Learn these specifics cold:

Their pre-feasibility study was written largely by their own employees, not independent experts. Their "zero waste" claim excludes the sediment-filled water they discharge back into the ocean. Greenpeace, the Environmental Justice Foundation, MiningWatch, and the International Seabed

Authority (the UN-linked regulator they bypassed by going through a U.S. permit instead) have all publicly attacked their credibility.

And distrust is expensive: TMC lost roughly **\$320M in 2025**, analysts doubt the project can ever be profitable, and ESG-screened investors and insurers won't touch claims they can't verify. **The trust deficit and the cash deficit are the same problem.** That's your key line: *TMC doesn't have a mining problem — it has a proof problem, and the proof problem is also the money problem.*

Your solution: Glass Ocean

Environmental sensors on the seafloor and in the water column (turbidity, water chemistry, whale-detecting microphones, biodiversity photo surveys) feed data to the ship. From the moment it arrives, the data is: **(1) signed and hash-chained in Splunk** — think of a record book where every page's fingerprint is stamped onto the next page, so ripping out or editing any page is instantly detectable; **(2) architecturally one-way** — the network is physically designed so data flows OUT of the sensor zone but nothing (not even TMC's own operators) can write INTO it; **(3) published on a public dashboard** regulators, scientists, and even Greenpeace can watch near-live.

The reframe that wins Innovation points: every other team will present security as *protection*. We present security as *the product* — the thing that makes TMC licensable, insurable, and fundable. Instead of "trust us," the ocean testifies for itself.

Your Cisco products (know what each does in one sentence)

Splunk (Cisco-owned — most teams won't know that; say it) = the tamper-evident record book. **Catalyst Industrial Ethernet** = rugged shipboard network carrying the data.

ThousandEyes = watches the ship-to-shore satellite path so an outage can be proven as fault vs. foul play. **Webex** = the "regulator observation room," a standing live window for NOAA and independent scientists.

Your demo: the website

The Glass Ocean site IS our centerpiece demo: a live-style dashboard (plume levels, whale detections, harvest decisions today), a **"Verify this data" button** that visibly runs the tamper-check — that's the money moment — a regulator-vs-public view toggle, and a data-governance page. The presentation concept opens on YOUR visualization: looking down

through a glass bridge into the ocean. You two own that opening image; rehearse the handoff from it into your talking points.

Questions you two will get (with answers)

"Couldn't TMC just turn the sensors off?" — Gaps are themselves visible in the ledger; missing data is an alarm, not a hiding place. *"Why would TMC pay for this?"* — Distrust is why they can't raise money; Glass Ocean is pitched as the key that unlocks capital, and Phase 0 costs a rounding error. *"Doesn't transparency hand critics ammunition?"* — Yes, and that's the point: only a company intending to operate within limits adopts this, and predictive monitoring warns TMC before a violation, not after. *"Isn't helping TMC unethical?"* — We're not defending seabed mining; we're making it impossible to do in the dark. If the verified data shows unacceptable harm, that evidence empowers regulators to stop it. Acknowledge the counterargument (transparency can legitimize a harmful activity) and say the team debated it — visible ethical reasoning scores.

Your rubric weight

You carry **Value + Ethics** as named categories, but also the heart of Innovation (30 pts), because Glass Ocean is the original idea. You have the highest-stakes slot: rehearse the most.

mining

Need to include: why are we implementing AI?

Problem: Outdated Mining Technology & Environmental Compliance

TMC's current mining vehicles cannot distinguish marine life from mineral nodules, causing unnecessary disturbance to deep-sea ecosystems. At the same time, regulators require accurate environmental data to prove mining operations remain within environmental limits. If this data is inaccurate, manipulated, or incomplete, TMC could lose regulatory approval.

Solution: AI-Powered Selective Harvesting + Environmental Sentinel Network

Retrofit TMC's existing mining vehicles with **AI-powered computer vision** that identifies marine organisms and selectively collects only safe nodules, reducing habitat disturbance and sediment plumes. Surrounding the mining operation, an **Environmental Sentinel Network** of underwater sensors continuously monitors water quality, sediment movement, ocean currents, biodiversity, and marine mammals.

AI analyzes both the mining vehicles and environmental sensors in real time to:

- Identify marine organisms and avoid sensitive habitats.
- Predict sediment plume movement before environmental thresholds are exceeded.
- Detect marine mammals and recommend slowing or pausing operations.
- Identify faulty or tampered sensors.
- Automatically alert operators when environmental conditions require action.

Verified environmental data is then published to **Glass Ocean**, creating a transparent record for regulators, scientists, and the public.

Cisco Technologies

- **Cisco UCS / AI Infrastructure** runs AI models onboard the vessel for real-time decision-making.
- **Cisco Industrial Ethernet** securely connects cameras, sensors, and mining equipment.
- **Cisco Cyber Vision** continuously monitors operational technology devices and detects abnormal behavior or compromised sensors.
- **Cisco Splunk** creates a tamper-evident audit trail, preserving the chain of custody for all environmental data.
- **Cisco XDR** detects cyber threats or attempts to manipulate environmental data.

- **Cisco ThousandEyes** monitors communication between the vessel and shore, ensuring environmental data reaches regulators reliably.

TEAM BRIEF 2 of 3 — AI SELECTIVE HARVESTING (Outdated Technology)

For the two presenters owning Problem/Solution 2

Project Glass Ocean — Bat Bridge Partners — Cisco HS Capstone 2026

The whole project in 90 seconds (same in all three briefs)

Our client is **The Metals Company (TMC)** — a real, publicly traded company trying to become the first commercial deep-sea miner. They harvest potato-sized rocks (polymetallic nodules) from the Pacific seafloor at 4,000m depth; the rocks are rich in nickel, cobalt, copper, and manganese — the metals batteries and data centers need. TMC is betting its future on AI-driven modernization, which is exactly the client the capstone prompt describes: excited about AI, worried about the security and ethics risks that come with it.

Our proposal has three linked parts: (1) **Glass Ocean** — a tamper-proof, publicly verifiable environmental monitoring platform, (2) **AI Selective Harvesting** — computer vision retrofitted onto their mining vehicles (your part), and (3) **Zero Trust Security** — the Cisco architecture protecting it all. The one-line thesis: *give the machines eyes, give the ocean a voice, and lock both behind doors nobody can pick.*

Scalability extensions (mention only if asked): SafeDepth (crew safety platform) and Abyssal Weather (ocean-current data for science and shipping).

Your problem: blind and mute machines

TMC's collector vehicles are conventional — tracked machines with suction systems whose basic design has barely changed since the 1960s. Two failures in one hull:

Blind: the machines cannot tell a bare nodule from one hosting living organisms (sponges, corals, strange single-celled giants called xenophyophores), and they kick up large sediment plumes — the two specific harms critics cite most.

Mute: the machines produce no trustworthy record of what they did. Even when TMC operates cleanly, they can't prove it — and their NOAA license depends on environmental compliance data. A single disputed dataset could suspend operations. No deep-sea miner today treats operational telemetry as evidence.

The opportunity hiding inside this: TMC's AI maturity is near zero, so **everything we propose is additive — nothing gets ripped out**. That's a rubric advantage: we never have to argue Cisco beats their existing AI, because they don't have any.

Your solution: the retrofit

Cameras and LED lighting mounted on the **existing** collector vehicles feed an onboard computer-vision model that classifies nodules in the vehicle's path in real time: bare nodule → harvest; nodule with visible life → skip, adjusting the collector head on the fly. A second model watches sediment sensors and automatically slows the vehicle when the plume gets too dense. Best analogy for judges: **adding driver-assist to a truck fleet the company already owns**.

Then the twist that connects you to the Glass Ocean: every decision the vehicle makes — creature avoided, speed reduced, threshold triggered — is **signed at the edge and written into the same tamper-proof ledger**. The machines get eyes AND a voice. The retrofit and the regulatory fix are one deployment, one budget.

Deployment honesty (this is your Sustainability material): the AI runs **on the ship, at the edge** — a Cisco UCS "AI POD" class server — because satellite bandwidth is nowhere near enough to stream seafloor video to a cloud, and a 700ms satellite round-trip is useless for split-second vehicle decisions. We fine-tune pre-trained vision models on nodule imagery rather than training from scratch (cheaper, greener, faster), and heavy retraining happens onshore between voyages.

Credibility anchor: cite **Impossible Metals** — a real startup whose robots already do vision-based selective nodule picking — as proof the concept works, then differentiate: they built a brand-new robot; we retrofit intelligence onto proven heavy equipment a real company already owns. Phasing: advisory mode first (AI recommends, human approves), automatic control only in a later phase — never promise unverifiable performance numbers.

Your Cisco products (one sentence each)

Cisco UCS (AI POD-class) = the shipboard AI brain running inference locally. **CURWB (Ultra-Reliable Wireless Backhaul)** = wireless links between vessel, support boats, buoys, and

surface drones — its real designed use case (ports, mines, moving industrial assets). **Catalyst Industrial Ethernet** = carries camera feeds and decision logs across the ship.

The honest note you must volunteer before judges ask: radio doesn't propagate underwater — the vehicle-to-surface leg uses acoustic modems or tethers, which are NOT Cisco products. Cisco takes over the moment data reaches a surface asset. Saying this unprompted converts a weakness into feasibility credibility. Do not skip it.

Your demo: the interactive one (the only interactive demo we have)

Sequence: the audience first sees the robot from **outside** — a 3D view with callouts labeling the added Cisco tech (camera pod, edge compute link, signed-telemetry module). Then the view drops into **first person**: you're the robot, scanning the seafloor — green box = bare nodule (collect), red box = life detected (skip) — and the audience member picks up a nodule. The simulation freezes on that moment and the solution slide appears behind it; that's your cue to start talking. Laptops circulate one at a time afterward for hands-on. You two own the demo handoff choreography — rehearse the freeze-to-talking transition until it's seamless.

Questions you two will get (with answers)

"Can cameras really work at 4,000m?" — Yes: offshore inspection ROVs run computer vision routinely today, and Impossible Metals classifies nodules with vision now; darkness is solved with LED arrays. *"How accurate is the classification?"* — We deliberately don't quote a made-up number; pilot phase establishes recall targets (e.g., >90% fauna detection) before automatic mode. *"Why not just buy Impossible Metals' robots?"* — Fleet replacement costs far more than retrofit, and TMC's capital position makes retrofit the only realistic path; also their robots are still pre-commercial. *"Does slowing for plumes hurt output?"* — Slightly, per hour — but one license suspension costs infinitely more than throttling; compliance IS the business case.

Your rubric weight

You carry **Technical + Sustainability** as named categories, plus the "AI Integration" line inside Technical Feasibility (25 pts). You also hold the demo most likely to make judges smile — interactive beats passive. Rehearse the tech handoffs; a fumbled demo costs Presentation points that clean slides can't buy back.

1. The Problem

Who: TMC (The Metals Company) — mines polymetallic nodules (potato-sized rocks rich in nickel, cobalt, manganese) from the deep seafloor.

What's wrong today:

- Their collector vehicles are "blind." They scrape/vacuum the seafloor for nodules but can't distinguish a nodule from a sponge, coral, or small marine animal sitting near/on it.
- This causes unnecessary habitat destruction and kicks up sediment plumes (clouds of stirred-up seafloor mud) that can smother marine life far from the collection site.
- Regulators (like the International Seabed Authority) require proof that mining stays within environmental limits. If TMC can't produce trustworthy data — or if data looks like it *could* have been altered — they risk losing their license to operate.

The two-part problem:

1. **Physical problem:** the machine can't see what it's damaging.
2. **Trust problem:** even if TMC operates cleanly, they currently have no bulletproof way to *prove* it.

2. The Solution — Plain English Version

Two systems working together:

A) AI Selective Harvesting (on the vehicle)

- Retrofit cameras + computer vision onto the *existing* mining vehicles
- The AI looks at the camera feed in real time and asks: "Is this a nodule, or is something living here?" This model runs on **Cisco UCS AI Infrastructure** installed aboard the vessel — the onboard "brain" powerful enough to make these calls in milliseconds, without needing to send video to the cloud.
- If it detects an organism, the vehicle skips that spot or slows down.
- If sediment plume levels start climbing too high, it slows or pauses automatically.
- Every single decision (avoided a creature, reduced speed, skipped an area) gets logged automatically into a secure, tamper-proof record via **Cisco Splunk**.

B) Environmental Sentinel Network (around the mining site)

- A network of underwater sensors watching the *surrounding* environment: water quality, sediment movement, currents, biodiversity, marine mammal presence.
- AI continuously analyzes all of this alongside the vehicle data to catch problems the vehicle itself might miss (e.g., a whale approaching from outside camera range).
- **Cisco Cyber Vision** continuously watches every sensor and camera on the network, so it can detect if a sensor itself is broken, malfunctioning, or has been tampered with — protecting the *integrity* of the monitoring system, not just the ocean.
 - Cisco Cyber Vision watches all the industrial equipment on a network and learns what "normal" behavior looks like for each device. If something deviates, it flags it in real time. For your project, it's the checkpoint that verifies your underwater sensors' data is trustworthy *before* it gets permanently logged by Splunk.

C) Glass Ocean (the trust layer)

- All verified data — video decisions, sensor readings, alerts — gets published to "Glass Ocean," a transparent public/regulator-facing record, built on the tamper-evident log **Cisco Splunk** maintains.
- This is the "voice" that proves the "eyes" (AI vision) actually did their job.

Why AI?

AI can learn to tell a living organism apart from a nodule in murky, variable seafloor footage. Non-AI systems (thresholds, basic motion detection) would either miss organisms or trigger constant false stops, since they can't distinguish subtle visual differences like camouflage or shape. AI also improves over time as it sees more footage, while rule-based systems stay static and need manual reprogramming for every new edge case.

cybersecurity

TEAM BRIEF 3 of 3 — ZERO TRUST SECURITY (Cybersecurity & Operational Risk)

For the two presenters owning Problem/Solution 3

Project Glass Ocean — Bat Bridge Partners — Cisco HS Capstone 2026

The whole project in 90 seconds (same in all three briefs)

Our client is **The Metals Company (TMC)** — a real, publicly traded company trying to become the first commercial deep-sea miner. They harvest potato-sized rocks (polymetallic nodules) rich in nickel, cobalt, copper, and manganese from the Pacific seafloor at 4,000m. TMC is betting its future on AI-driven modernization — exactly the client the capstone prompt describes: excited about AI, worried the new security and ethics risks could let hackers break in.

Our proposal has three linked parts: (1) **Glass Ocean** — a tamper-proof, publicly verifiable environmental monitoring platform, (2) **AI Selective Harvesting** — computer vision retrofitted onto mining vehicles, and (3) **Zero Trust Security** — the Cisco architecture protecting it all (your part). One-line thesis: *give the machines eyes, give the ocean a voice, and lock both behind doors nobody can pick*. You are the "doors nobody can pick" team.

Scalability extensions (mention only if asked): SafeDepth (crew safety platform on the same network) and Abyssal Weather (ocean-current data products).

Your problem: a floating industrial plant with a target on its back

TMC's operation is a huge, exposed attack surface made almost entirely of **operational technology (OT)** — the computers that control physical machines: a production vessel that is essentially a floating factory, remotely operated and autonomous underwater vehicles, an

environmental sensor network whose data feeds regulatory decisions, and a single satellite link connecting it all to shore.

And TMC is unusually *motivated* a target: it operates under a U.S. permit that the UN-linked International Seabed Authority calls a violation of international law, meaning nation-states, activist groups, and competitors all have reasons to disrupt, discredit, or spy on the operation.

Memorize the three nightmare scenarios — they make the threat concrete:

1. **Data poisoning:** a hostile actor corrupts environmental monitoring data — either faking a violation (license suspended) or hiding a real one (credibility destroyed when discovered). Because the Glass Ocean's whole value is trust, an attack on the data IS an attack on the company.
2. **Ransomware at sea:** the ship's industrial controls are locked mid-operation, hundreds of miles from help. Land-based analogy you'll cite: the Ascension hospital attack — one malicious download, and because nothing was segmented, 7 compromised servers took down a 25,000-server network for six weeks.
3. **Vessel spoofing:** AIS (ship-tracking) manipulation near the operating area creates collision risk or false narratives — a lesson from the 2024–25 Baltic Sea cable incidents, where attribution proved nearly impossible.

Your solution: Zero Trust architecture

"Zero Trust" in one sentence: **nobody and nothing is trusted by default — every person, device, and connection must prove itself, every time.** Four design moves:

1. Identity first (Duo + Secure Access): phishing-resistant multi-factor authentication for every human and every remote session. Real-world anchors: MGM was breached by one phone call to a help desk; Colonial Pipeline fell to one password on an account with no MFA. Shore engineers connect through zero-trust network access (ZTNA), not an old-style flat VPN.

2. Segmentation (ISE + industrial firewalls): the ship's network is divided into four sealed zones — crew IT, mining controls, environmental sensors, satellite uplink. A hacked crew laptop can never reach a collector vehicle's controls. This is the direct anti-Ascension design.

3. The one-way rule (data diode pattern): the environmental sensor zone is architecturally exit-only — data flows out to the ledger and dashboard, but nothing on the ship, including TMC's own operators, can write into it. This single design choice is what upgrades "TMC can't fake its data" from a policy promise to a **provable architectural fact** — and it's the load-bearing wall under the entire Glass Ocean pitch. When the Problem 1 team says "tamper-proof," YOUR architecture is why that word is true.

4. Watch everything (Cyber Vision + Umbrella + XDR + Splunk): Cyber Vision passively fingerprints every industrial device and flags abnormal machine commands; Umbrella blocks

malicious downloads at the DNS layer (the exact Ascension entry vector); XDR correlates alerts across endpoint, network, and identity into one picture; Splunk (a Cisco company) keeps the immutable audit log for both security events and environmental data.

Your Cisco products (one sentence each — you have the most)

Duo = everyone proves who they are. **Cisco Secure Access (ZTNA)** = zero-trust remote entry for shore staff. **ISE** = enforces the ship's sealed zones. **Cyber Vision** = security camera for machine behavior. **Umbrella** = blocks bad sites/downloads at DNS. **XDR** = the correlation brain. **Splunk** = the tamper-evident record book (shared with Team 1 — coordinate so you don't both explain it; suggested split: they explain WHAT it proves, you explain HOW it can't be tampered with).

Your demo: the shark attack (passive visualization)

Concept: a stylized visualization where shark attacks represent cyberattacks on the operation — a phishing shark probing the crew network, a ransomware shark lunging at mining controls, a data-poisoning shark circling the sensor zone — each one visibly stopped by a labeled Cisco defense layer (Duo gate, ISE walls, the one-way valve). Not interactive; it plays as a story while you narrate. Your job is writing tight 15–20 second narration beats matching each shark to its real-world incident (MGM, Ascension, Baltic). It's a metaphor, so land it with one self-aware line ("the sharks down here don't have fins") and move on — don't let the visual outshine the architecture.

Questions you two will get (with answers)

"Isn't Zero Trust overkill for a mining ship?" — The ship is a floating industrial plant whose data determines its license; Ascension shows what flat networks cost (\$1.1B, six weeks of paper charts). *"What if the satellite link itself is cut?"* — Edge autonomy: the AI and controls run locally on the ship; the link carries telemetry, not decisions, and ThousandEyes proves whether an outage was fault or attack. *"Can't insiders bypass all this?"* — The one-way design is specifically insider-proof: even administrators can't rewrite the ledger, and gaps in data are themselves visible alarms. *"Why Cisco instead of any security vendor?"* — Breadth in one integrated stack: identity, industrial OT monitoring, DNS, correlation, and logging that natively interoperate — plus Cisco's IoT/industrial portfolio (CURWB, Catalyst IE) already runs the operational network, so security and networking are one architecture, not bolted together.

Your rubric weight

You carry the **Security & Ethics pillar's** technical half and much of **Technical Feasibility (25 pts)** — you name more real Cisco products than both other teams combined, which is where "Use of Cisco Technologies" points live. You're also the pair most likely to face the technical coach's hardest questions: know WHY each product exists, not just its name. If you can explain the one-way data design clearly, the whole project's credibility holds.